

MOOSA AHMED

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PROFESSIONAL SUMMARY

Final-year Computer Science student with project experience in software development, relational database design, reinforcement learning, and UX/UI design. Seeking an entry-level software development role.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript, C#

Technologies: .NET, WPF, PostgreSQL, MySQL, Entity Framework Core, Git, Linux

Concepts: Object-Oriented Programming, Data Structures & Algorithms, Database Design, Reinforcement Learning, Agile/Scrum

PROJECTS

Smart Parking System | Python, Q-Learning, Dijkstra's Algorithm, Matplotlib Winter 2026

- Implemented **Q-learning** from scratch in **Python** to model dynamic parking selection as a **Markov Decision Process** without a reinforcement learning library.
- Compared the approach with a **Dijkstra** and **hill-climbing** baseline across 3 constraint scenarios, visualizing reward convergence with **Matplotlib**.
- Co-authored an 8-page IEEE-format paper with a 4-person team, communicating the optimization and adaptability trade-off to a non-technical audience.

Car Rental System | C#, .NET, WPF, PostgreSQL, Entity Framework Core Fall 2025

- Built a **WPF** desktop application supporting account registration, profile setup, and car-listing browsing by year, make, model, location, and price.
- Designed a 4-entity **PostgreSQL** database and integrated it with the application using **Entity Framework Core**, **Npgsql**, and version-controlled migrations.
- Served as **Scrum Master** for a 7-person **Agile** team, coordinating sprint planning and backlog grooming across 3 sprints and 2 milestones.

Equipment Test Center DBMS | MySQL, Python, Tkinter Fall 2025

- Designed and normalized an 11-entity relational database, applying **ER modeling** and decomposition through **3NF/BCNF** with primary and foreign-key constraints.
- Wrote 10+ analytical **SQL** queries, including **correlated subqueries** and **window functions** such as **RANK**, **ROW_NUMBER**, and **NTILE**.
- Built a **Python/Tkinter** reporting GUI connected to **MySQL** as part of a 2-person team across the database design lifecycle.

EXPERIENCE

Backup Supervisor & Storefront Associate | Tim Hortons Nov 2024 – Present

- Took on shift-leadership responsibility as Backup Supervisor, learning to delegate, stay calm under pressure, and keep a team accountable in a fast-paced environment.
- Built consistency and attention to detail handling cash and customer concerns independently, while holding myself to food-safety and cleanliness standards without oversight.

Peer Educator | Wilfrid Laurier University Jan 2024 – May 2024

- Developed my communication and patience facilitating academic support sessions for international students, learning to adapt explanations to different backgrounds and comfort levels.
- Strengthened my reliability and follow-through by collaborating with faculty over a full semester to prepare and deliver materials for a diverse student cohort.

EDUCATION

Honours Bachelor of Science in Computer Science | Wilfrid Laurier University | Expected Dec 2026

Relevant Coursework: Data Structures & Algorithms, Object Oriented Programming, Database Systems, Software Engineering, Computer Networks, Artificial Intelligence, Machine Learning, Data Mining, Information Retrieval